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Research Article

Spider-silk-inspired dynamic poly(urethane-urea) networks with mechanical reinforcement via mismatched supramolecular interactions for luminescent fibers

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ABSTRACT

Most current soft and flexible luminescent matrices exhibit poor mechanical performance, leading to limited performance reliability and longevity. Inspired by the heterogeneous structure of spider silk, we developed a sacrificial, robust, and self-healing poly(urethane-urea) (Supra-PU) elastomer with a physically hydrogen-bonded crosslinked network via the mismatched mechanistic interplay between flexible and rigid segments. This engineered structure leads to denser hard domains and effective energy dissipation, enabling the optimal Supra-PU elastomer to exhibit extraordinary mechanical properties, such as an ultrahigh strength of 30.6 MPa with a fracture strain of 1251.8 %, a true stress at break of 423.3 MPa, toughness of 102.9 MJ m^{-3}, and damage-tolerant performance with a high fracture energy of 4.7 kJ m^{-2}. Moreover, loosely packed and noncrystallized hard domains impart the desired network with low chain restriction and high relaxation dynamics, resulting in a high healing efficiency of 92.1 % with restored toughness of 94.8 MJ m^{-3}. Furthermore, luminescent fibers are achieved via wet spinning with superior mechanical properties and cyan afterglow (~0.8 s). This meticulous molecular engineering provides valuable insights for advancing high-performance self-healable poly(urethane-urea) and luminescent materials.

1. Introduction

Newly emerging luminous materials have garnered widespread interest for their potential application in displays, bioimaging, wearable electronics, and encryption fields [1–4]. Various feasible strategies have been exploited to impart flexible artificial materials with this fascinating function. Terpyridine phenylboronic acid organic phosphor-embedded poly(vinyl alcohol) composites have been developed with phosphorescence for information encryption [5]. More recently, a bimodal sensor was designed by incorporating copper-doped ZnS nanoparticles into a polydimethylsiloxane matrix [6]. Many polymeric materials, including poly(vinylidene fluoride-trifluoroethylamine) [7], epoxy resin [8], and silicone [9], have been utilized as matrices to provide a large space for the design of luminescent devices. However, long-term operation, repeated stretching stimuli, and external mechanical damage during practical applications may lead to their degradation and functional failure. Conventional polymer matrices do not exhibit self-repair attributes due to their limited chain mobility and lack of bond reformation abilities [10,11]. Therefore, the development of luminescent materials still faces challenges.

Polyurethanes (PURs) possess satisfactory flexibility and durability that qualify them as high-performance polymeric materials. Studies on self-healing PURs emerged a couple of decades ago and continue to attract the scientific community [10,12,13]. The ability to self-heal can repair external or internal damage, thereby ensuring material performance reliability and lifetime [14–16]. The first report on heterogeneous PUR chemistries was introduced in 2009, with a self-repairing network based on oxetane-substituted chitosan [17], followed by several strategies proposed to construct sacrificial networks via intra/intermolecular interactions for the healing design, including supramolecular systems [18–20], ionic interactions [21,22], coordination bonds [23–25], reversible covalent bonds, etc. [11,26–29]. Despite significant advancements achieved in self-healing PURs, most of them depend on external conditions to initiate chain exchange for the healing process, and their appli-

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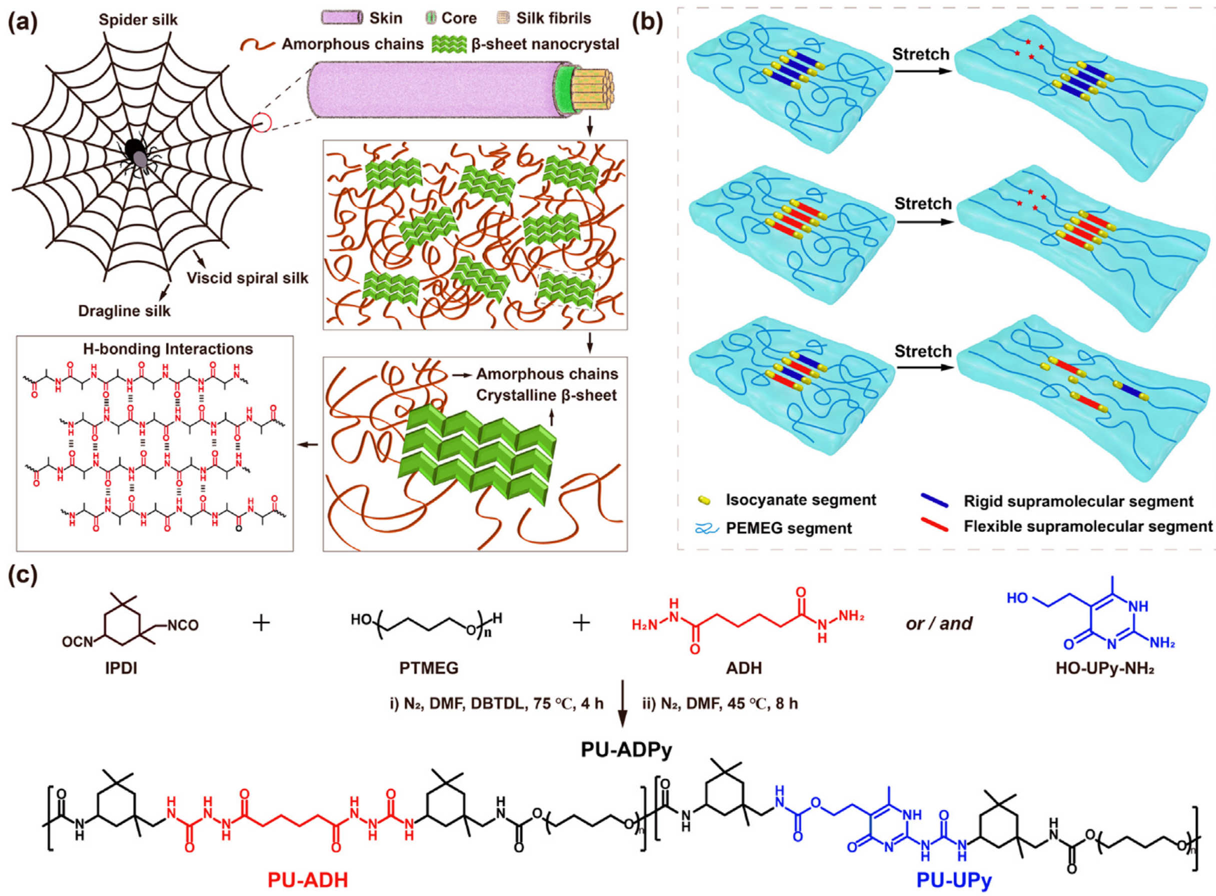

Fig. 1. (a) Schematic illustration of the spider silk molecular structure. (b) Cooperative rigid and flexible supramolecular segments for toughened elastomers. (c) Synthesis routes of poly(urethane-urea) elastomers with different chain extenders.

cations currently remain limited due to the conflicting attributes of strength versus mobility [13]. Simultaneously achieving polymeric materials with excellent mechanical strength and autonomic ambient temperature self-healing remains a challenge.

Numerous natural materials exhibit efficient self-repair with an extraordinary combination of contradictory mechanical performance [30,31]. Their unique hierarchical structural features, spanning multiple length scales from the molecular to the near-macroscopic dimensions, inspire researchers to develop advanced self-healable materials [32]. With a two-phase structure, spider silk has ultrahigh stiffness (3–10 GPa) and extensibility (15 %–45 %) [33]. A theoretical investigation revealed that beta-nanocrystalline regions separated by amorphous linkages with dense hydrogen bonding interactions are the reason for the superb toughness of spider silks (Fig. 1(a)) [34,35]. Supramolecular chemistry exhibits attractive features in self-healing: reversibility, directionality, and sensitivity [14]. Therefore, reasonable bionic structural design with a noncovalent crosslinked network may be an effective strategy for the development of high-performance and autonomic ambient self-healing elastomers [36]. 2-Ureido-4[1H]-pyrimidinone (UPy) was first synthesized by Meijer and colleagues [37] as functionalized groups for reversible supramolecular polymers. A self-healing and robust elastomer was synthesized by combining quadruple H-bonding and polyrotaxanes dual crosslinks [38]. However, fixed dimerization results in an upper limit on supramolecular toughening, whereas nonspecific supramolecular building blocks are inefficient in dissipating energy due to excessive supramolecular aggregation under external stresses (Fig. 1(b)) [39]. In most cases, the mechanical performance of these materials still suffers from limited enhancement (generally lower than 20 MPa) owing to the weakness of noncovalent bonds and the balance between energy dissipation and supramolecular bonding interactions.

Inspired by the structure of spider silk, we developed a mechanically robust, highly stretchable, and room-temperature self-healing supramolecular poly(urethane-urea) network (Supra-PU) via the mechanistic interplay between flexible and rigid chain extenders for toughening luminescent fibers. The incorporation of functional segments with differing structural rigidity engenders mismatched supramolecular interactions for the physically H-bonded crosslinked poly(urethane-urea) network, which can efficiently tune energy dissipation and enhance load-bearing capacity. For demonstration, various Supra-PU elastomers have been synthesized, and the optimal supramolecular elastomers containing flexible and rigid segments (PU-AD_1Py_1) exhibit significantly enhanced mechanical properties compared to those with only single-type segments, with greater toughness of 102.9 MJ m^{-3}, super true fracture stress of ~423.3 MPa, good extensibility of 1251.8 %, and room-temperature self-healing performance. The numerous H-bond acceptors and donors contained in hard domains of poly(urethane-urea) networks are constructed via multiple acylsemicarbazide, urethane, and urea, and UPy units connected with asymmetric alicyclic spacers (isophorone diisocyanates (IPDIs)), resulting in noncrystallized and loose structures with high-density noncovalent hydrogen-bonding interactions and efficient chain mobility [40]. Therefore, the synthesized PU-AD_1Py_1 exhibits a superior tensile strength, high toughness, and excellent fracture strain. Denser hydrogen bonding-based hard regions coupled with a lower crystallization energy barrier of soft segments (poly(tetramethylene oxide) (PTMEG)) facilitate fast polymer chain diffusion and intermolecular bond reorganization, resulting in an efficient self-healing capability at room temperature [19]. Furthermore, the prospective application of poly(urethane-urea) luminescent fibers (Cu-ZnS/PU-AD_1Py_1) via wet spinning in light-emitting fields has been tentatively investigated.

2. Results and discussion

2.1. Synthesis and characterization of poly(urethane-urea) elastomers

The specific synthetic route of various Supra-PU networks is depicted in Fig. 1(c), and the detailed compositions are shown in Table S1 in the Supplementary Materials. Following the procedure reported by Wang et al. [41], we initially synthesized a rigid chain extender (OH-UPy-NH_2) via the reaction of α-acetylbutyrolactone and guanidine carbonate. The Fourier transform infrared (FT-IR) spectra (Fig. S1) of each monomer and the reactant OH-UPy-NH_2 are shown. ^{1}H NMR (Fig. S2) and HRMS (Fig. S3) were employed further to confirm the chemical structure. IPDI, PTMEG, the flexible chain extender (ADH), and/or the rigid chain extender (OH-UPy-NH_2) were reacted in the presence of DBTDL. At an R-value ([NCO]/[OH and/or NH_2]) of 1.14, the isocyanate groups from IPDI react with hydroxyl groups from polyols and amino groups from chain extenders AD and OH-UPy-NH_2, which respectively introduce the urethane and urea linkages into the Supra-PU networks. The molecular network structures of the Supra-PU elastomers are presented in Fig. 1(c): PU-UPy, with only rigid segments; PU-ADH, with only flexible segments; and PU-ADPy, with both above segments.

The chemical structures of poly(urethane-urea) elastomers were confirmed by the FT-IR spectra (Figs. S4–S6) and ^{1}H NMR analysis (Figs. S7–S11). The disappearance of the characteristic absorption peak at 2264 cm^{-1} during the reaction served as direct evidence for the complete consumption of isocyanate groups, thereby ensuring the safety of the poly(urethane-urea) films for application. As shown in Fig. S12, the synthesized Supra-PU polymers possess number-average molecular weights (Mn) ranging from 7.4 k to 23.0 k g mol^{-1} with polydispersities between 1.73 and 2.15. The prepared PU-ADH, PU-UPy, and PU-AD_1Py_1 films are highly transparent compared with PU-AD_1Py_2 and PU-AD_2Py_1, with transmittances > 80 % across the visible region (Fig. S13). It has been demonstrated that Supra-PU elastomers exhibit a similar low glass transition temperature of approximately −65 °C (Fig. S14), and good thermal stability ($T_{d,\,5\,\%}$ = 261.7–307.6 °C, Fig. S15, Table S2). Generally, the intrinsic self-healing of polymers primarily relies on chain interdiffusion above T_g and the reversible cleavage and reformation of dynamic bonds [42]. Furthermore, high-density hydrogen bonding interactions within polymer networks may lead to crystallization or clustering, making them brittle with low mechanical performance [43]. Differential scanning calorimetry (DSC) analysis was performed on the second heating cycle to investigate the crystalline structure of the poly(urethane-urea) polymers, as functional supramolecular segments with abundant H-bond sites are incorporated as hard domains in the networks. As shown in Fig. S14, the profiles exhibit no sharp crystallization and melting peaks in the temperature range from −80 to 180 °C, indicating the Supra-PU elastomers are amorphous. This is further corroborated by X-ray powder diffraction (XRD) analysis (Fig. S16), which reveals only a broad halo centered at $2\theta = 20°$. This absence of long-range order confirms the amorphous structure of the Supra-PU elastomers, which facilitates the chain diffusion essential for self-healing. Supramolecular interactions play significant roles in mediating the elastomer properties. The initial degradation temperatures of PU-ADPy elastomers are between those of PU-ADH and PU-UPy, illustrating that the mismatched supramolecular segments may affect the thermal stability of Supra-PU elastomers (Fig. S15, Table S2). The peaks of the carbonyl (C=O) region (1600–1750 cm^{-1}) were further analyzed by FT-IR analysis [44], which shows that the fractions of H-bonded C=O follow the sequence of PU-AD_1Py_1 (60.5 %), PU-UPy (59.5 %), and PU-ADH (55.5 %) (Fig. S17).

2.2. Mechanical performance of the Supra-PU elastomers

Toughness is determined by the strength and ductility of materials and is a crucial mechanical parameter for absorbing energy during plastic deformation and fracture [45]. As a result, we evaluated the mechanical performance of the Supra-PU elastomers at room temperature via tensile tests to obtain typical engineering stress-strain curves. As displayed in Fig. 2(a), PU-UPy has a low toughness of 3.5 MJ m^{-3}, with a tensile strength of 0.65 MPa and a fracture strain of 539.7 %. PU-ADH has an ultimate engineering strength of 17.1 MPa and a fracture strain of 1015.9 %, resulting in a toughness of 67.5 MJ m^{-3}. Comparatively, the ultimate engineering fracture stress of PU-AD_1Py_1 is as high as 30.6 MPa and remains at a fracture strain of ∼1251.8 %, showing the highest toughness of 102.9 MJ m^{-3}. Furthermore, PU-AD_2Py_1 and PU-AD_1Py_2 exhibit inferior mechanical performance (Fig. 2(b)). Specifically, PU-AD_2Py_1 has a tensile strength of 0.13 MPa and a fracture strain of 1033.7 %, while PU-AD_1Py_2 shows 1.1 MPa and 2539.8 %, respectively. Compared with PU-ADH, the PU-AD_2Py_1 and PU-AD_1Py_2 networks exhibit increased fracture strain and a marked decrease in tensile strength. Although mismatched supramolecular interactions between rigid and flexible chain extenders can improve the chain extensibility of poly(urethane-urea) networks, the mechanical properties of PU-ADPy films depend on the degree of microphase separation and the integrity of the physically cross-linked network. An imbalance in the molar ratio of rigid to flexible chain extenders may disrupt the optimal micro-separated morphology. This loss of nanoscale structural order compromises the effectiveness of hard domains as physical cross-linking junctions, resulting in an overall decline in mechanical performance. In the PU-AD_1Py_1 network, functional segments with mismatched supramolecular interactions in a balanced molar ratio efficiently regulate energy-dissipation pathways and improve resistance to external loading. Hidden lengths of the polymer chains were released under stretching via dynamic dissociation and reassociation of multiple reversible hydrogen bonds. The dissipation of substantial energy during both dissociation and reassociation results in a significant increase in toughness. The specific mechanical parameters are summarized in Fig. 2(e and f) and Table S3 to more intuitively reflect the mechanical performance changes. With respect to the effects of a decreased cross-sectional area on stress and a reduction in strain, true stress-strain curves were also calculated to illustrate their resistance to external stress (Fig. 2(c and d)) [39,46]. The results show that PU-AD_1Py_1 is significantly greater than the other elastomers, with the maximum true fracture stress of ∼423.3 MPa. A rectangular PU-AD_1Py_1 film (0.089 g, 5 mm in width, 0.4 mm in thickness) can withstand a load of 1.0 kg, which is ∼11,236 times its own weight, indicating the exceptional mechanical robustness of PU-AD_1Py_1 (inset in Fig. 2(c)).

Consecutive cyclic loading tests were performed on the Supra-PU elastomers to further elucidate their energy dissipation capabilities. Compared with other elastomers, PU-AD_1Py_1 networks with mismatched flexible and rigid segments present more prominent hysteresis loops for incremental applied strains, demonstrating the optimized energy dissipation capability (Fig. 2(g) and S18). Owing to its remarkable energy dissipation ability, the predamaged PU-AD_1Py_1 film shows high crack tolerance. As shown in Fig. 2(h and i), a notched PU-AD_1Py_1 sample can elongate to over 8 times its initial length, whereas the notch does not propagate further with the crack tip. According to the Greensmith method, the PU-AD_1Py_1

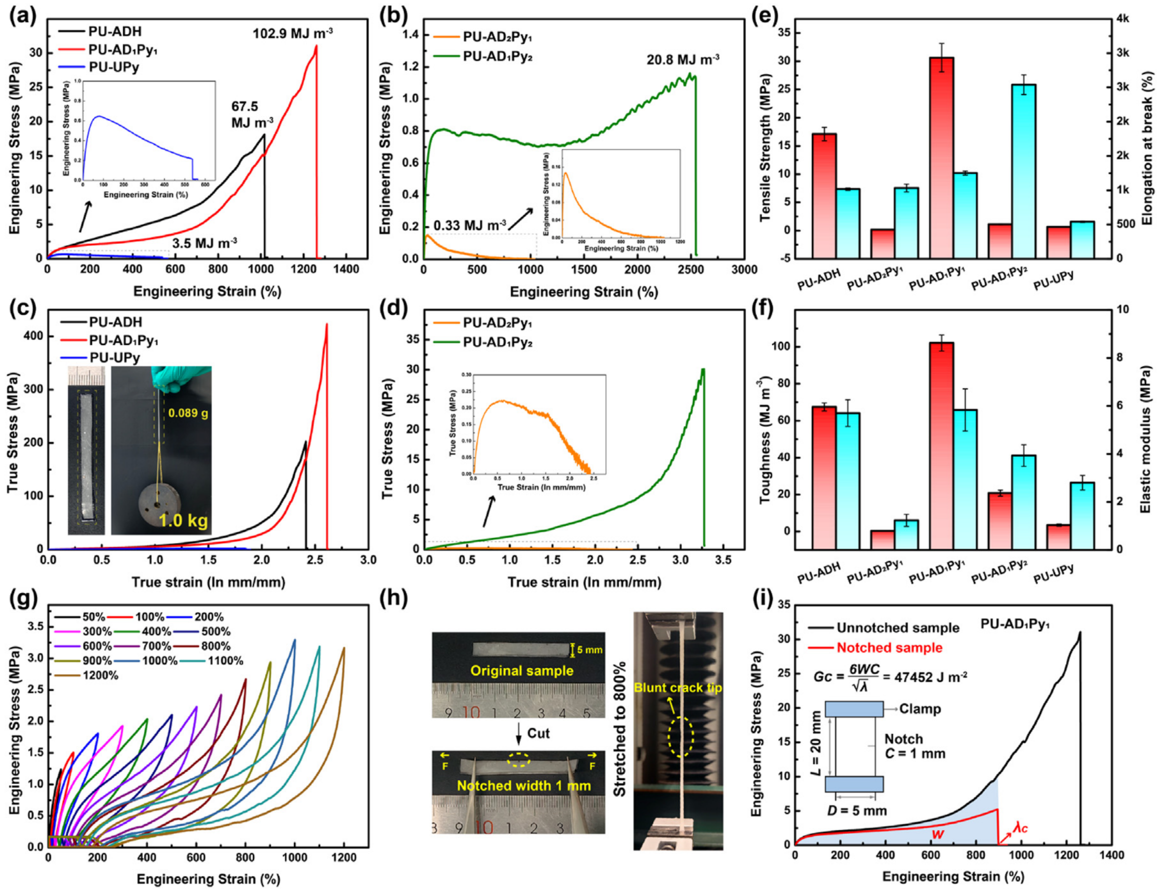

Fig. 2. (a) Stress-strain curves of PU-ADH, PU-AD_1Py_1, and PU-UPy, and (b) PU-AD_1Py_2 and PU-AD_2Py_1. (c) True stress-strain curves of PU-ADH, PU-AD_1Py_1, and PU-UPy, and (d) PU-AD_1Py_2 and PU-AD_2Py_1. Inset in (c) showing the photograph of the PU-AD_1Py_1 film (0.089 g) withstood a 1.0 kg load. (e) Tensile strength and fracture strain, and (f) toughness and Young's modulus of the poly(urethane-urea) elastomers. (g) Sequential cyclic tests with different applied strains of PU-AD_1Py_1. (h) Photographs of the original and notched PU-AD_1Py_1 films deformed to 800 % strain under a single load. (i) Tensile stress-strain curves of intact and notched PU-AD_1Py_1 films.

sample has a high fracture energy of 47,452 J m^{-2}. High-density H-bonds in hard domains with mismatched supramolecular interactions undoubtedly contributed to energy dissipation near the advancing crack tip, resulting in fracture stress transfer to the entire polymer network via crack bridging in the stick-slip motion of the polymer chains [23,47,48]. Hence, the PU-AD_1Py_1 sample can effectively avoid stress concentrations at the crack front, promoting high toughness and satisfactory crack tolerance.

2.3. Effect of supramolecular interactions

Supramolecular chemistry is crucial for developing mechanically tough and stretchable thermoplastic elastomers, and the mechanistic interplay of supramolecular segments is beneficial for constructing sacrificial and effective energy dissipation polymer networks [18,19,49]. A systematic study was conducted to explore the effect of supramolecular interactions on elastomer toughness. To isolate the contribution of the specific supramolecular segments, we disregard the same components of the Supra-PU elastomers (i.e., IPDI and PTMEG) and synthesize bis-ethyl isocyanate-modified ADH (Bis-EI-ADH) and bis-ethyl isocyanate-modified OH-UPy-NH_2 (Bis-EI-UPy) (Fig. S19), which allows us to investigate how these functional segments amplify the supramolecular effect and to better understand the contribution of supramolecular functionalities. The chemical structures of Bis-EI-ADH and Bis-EI-UPy were identified by proton nuclear magnetic resonance (^{1}H NMR, Fig. 3(a and b)). Thermogravimetric analysis indicates that Bis-EI-ADH and Bis-EI-UPy exhibit good thermal stability (Fig. S20). Then, various molar ratios of Bis-EI-ADH and Bis-EI-UPy were mixed together to further analyze the mechanism of mismatched supramolecular interactions. The arrangement between polymer chains may be influenced by the supramolecular interplay, giving rise to a significant effect on the thermal transition behavior of polymeric materials [39]. As depicted in Fig. 3(c), increasing Bis-EI-UPy content alters the original hydrogen-bonding network structure and average interaction strength of Bis-EI-ADH. This leads to changes in the movement capabilities of chain segments within its structure, resulting in the endothermic peak shifting toward a lower temperature.

Quantum chemistry calculations were performed on small-molecule models for further exploration of the binding energies of three dimers, i.e., those comprising rigid (Bis-EI-UPy) and/or flexible supramolecular segments (Bis-EI-ADH). The optimal conformations and interaction energies of the three dimers are shown in Fig. 3(d), revealing that hydrogen bonding and π-π interactions are the main driving forces between the supramolecular segments. Comparatively, the optimal conformation of ADH-UPy is more irregular than that of the ADH dimer and UPy dimer. The ADH dimer exhibits a stronger binding interaction than ADH-UPy and UPy dimer, with calculated binding energies (ΔE) of -153.39, -123.02, and -106.21 kJ mol^{-1}, respectively. The stronger binding affinity of ADH dimer could lead to greater aggregation of hard domains in PU-ADH, therefore resulting in more remarkable microphase separation than in PU-ADPy and PU-UPy, as evidenced by the small-angle X-ray scattering (SAXS) results in Fig. 4(a). Notably, stronger binding affinity for ADH dimer does not necessar-

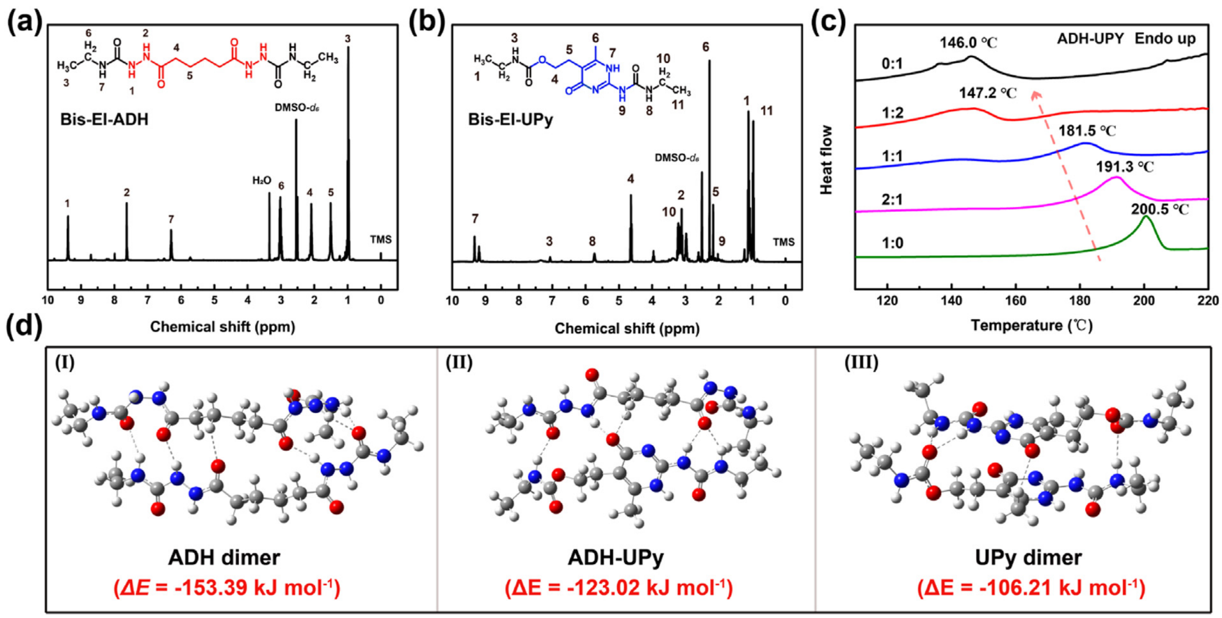

Fig. 3. Effect of supramolecular interactions. ^{1}H NMR spectra (400 Hz, DMSO–d_6) of (a) Bis-EI-ADH and (b) Bis-EI-UPy. (c) DSC curves. (d) Binding energies (ΔE) between various chain extender dimers (ADH dimer, ADH-UPy, and UPy dimer).

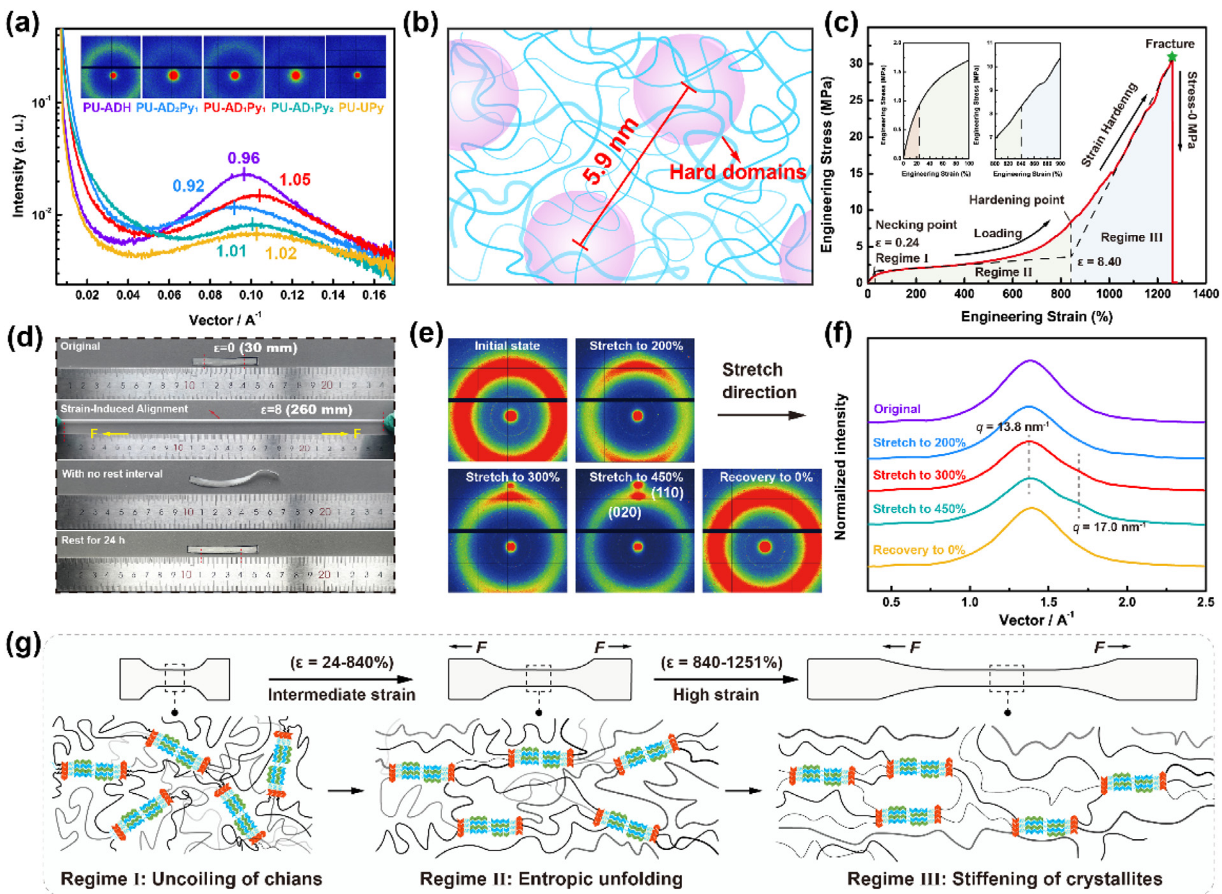

Fig. 4. Mechanical toughening mechanism in PU-AD$_1$Py$_1$ elastomers. (a) 1D SAXS profiles and 2D SAXS patterns of poly(urethane-urea) elastomers. (b) Schematic illustration of the microstructure of the PU-AD$_1$Py$_1$ network. (c) Dependence of the stress on the strain divided into three regimes. (d) Photographs displaying the recoverability of the original, stretched, and recovered PU-AD$_1$Py$_1$ films after 24 h of rest. (e) 2D patterns and (f) 1D Wide-angle X-ray diffraction (WAXD) profiles of the PU-AD$_1$Py$_1$ film at various applied strains. (g) Schematic description of the proposed toughening mechanism in PU-AD$_1$Py$_1$ during stretching.

ily equate to higher toughness in PU-ADH networks. Supramolecular segments with strong nonspecific binding may restrict energy dissipation, thereby compromising the poly(urethane-urea) networks' ability to withstand external forces [50]. The sliding of polymer chains is crucial for preventing local stress accumulation in elastomers, which can effectively dissipate external stress and maintain network integrity [39]. Therefore, rationally designing the supramolecular functionalities within elastomers to balance physical crosslinking strength and energy dissipation ability is essential for developing mechanically robust polymeric elastomers. With regard to the previous mechanical performance comparison of Supra-PU elastomers, mismatched interplay within polymer networks is the key to imparting superb stretchability and high toughness to PU-AD_1Py_1 elastomers.

2.4. Toughening mechanism of the PU-AD_1Py_1 elastomer

High-density H-bonding interactions induce Supra-PU chain aggregation with a phase separation in polymer networks. SAXS patterns were measured to explore the microphase separation structures of the Supra-PU elastomers. As shown in Fig. 4(a), all the Supra-PU elastomers display a wide scattering peak, and the sequence of peak intensity is as follows: PU-ADH>PU-AD_1Py_1>PU-AD_2Py_1>PU-AD_1Py_2>PU-UPy. The periodicity, which presents a hard domain average distance, is calculated to be 6.5, 6.8, 5.9, 6.2, and 6.1 nm for PU-ADH, PU-AD_2Py_1, PU-AD_1Py_1, PU-AD_1Py_2, and PU-UPy, respectively. Differences in the binding affinities of supramolecular segments result in variations in microphase separation within poly(urethane-urea) networks. Given the nearly identical volume fractions of hard domains for the Supra-PU elastomers, hard domains with the shorter distance in PU-AD_1Py_1 demonstrate the smaller average size within the crosslinked network. Specifically, the PU-AD_1Py_1 network has more dense hard domains with smaller sizes, and highly extensible soft chains are crosslinked (Fig. 4(b)). This morphology benefits Supra-PU elastomers. Hard domains with a higher density within polymer networks are beneficial for elastic restorability, and mismatched supramolecular interactions within physically crosslinked networks contribute to effective energy dissipation, leading to polymer networks' high ductility and toughness [46].

The mechanical behavior of PU-AD_1Py_1 was analyzed further to better explore the toughening mechanism. As shown in Fig. 4(c and g), the PU-AD_1Py_1 film presents a typical J-shaped stress-strain curve with three distinct mechanical response regimes. The initial linear response originates from the uncoiling and straightening of molecular chains (linear stage: 0–24 %, regime I). After the yielding point, the entropy of the soft segments within the amorphous network unfolds (yield stage: 24 %–840 %, regime II). Upon further increasing the applied strain, the load could be effectively dissipated to decrease the stress concentration, and the molecular chains could align and straighten alone in the tensile direction, leading to a prominent increase in the modulus (strain hardening stage: 840 %–1251 %, regime III) [36]. The variation in the strain-stress curves is likely correlated with molecular chain rearrangement, disentanglement, and alignment [19]. As displayed in Fig. 4(d), the initial rectangular PU-AD_1Py_1 film exhibits stress whitening when stretched to a high strain of ~800 %. Immediately upon unloading, the network contracts from its extended state, causing the sample to coil and slowly recover its original shape. After 24 h of relaxation, the sample nearly recovers to its initial state, demonstrating its exceptional elastic restorability. This phenomenon signifies that the initially amorphous PU-AD_1Py_1 network undergoes a phase transformation, accompanied by a change in refractive index during stretching. This transformation is likely attributable to the crystallization of soft segments within polymer networks, resembling the strain-induced crystallization (SIC) performance that occurs in natural rubber [51]. Such a process contributes to improving strength and fatigue resistance. 2D WAXD patterns and 1D WAXD profiles are then acquired by elongating the PU-AD_1Py_1 film with various applied strains and then releasing to verify the hypothesis (Fig. 4(e and f)). The initial pattern displays the isotropy of PU-AD_1Py_1 before it is stretched. Subsequently, the single isotropic scattering ring progressively weakens along the stretch direction, and the vertical scattering signal eventually increases in the perpendicular direction. There are distinguishable spots appearing in the patterns at strains of 300 % and 450 %, which correspond to the scattering vectors q = 13.8 nm^{-1} and q = 17.0 nm^{-1}, respectively, attributed to the (020) and (110) planes of the crystallized PTMEG soft segments [19]. Furthermore, the released film can return after restoration to the initial amorphous structure, as shown in the 1D curve and 2D pattern. These results illustrate that PTMEG chains undergo reversible strain-induced alignment and crystallization, which benefits the superior toughness and elasticity of polymers.

2.5. Self-healing performance of the PU-AD_1Py_1 elastomer

Supramolecular hydrogen bonding interactions are essential for constructing sacrificial and reversible polymer networks. As a reversible strategy, the key point of SIC is to provide enough regulatory space for achieving high chain mobility in hard domains and effective dissociation/association of dynamic hydrogen bonding interactions [19]. As previously mentioned, mismatched supramolecular interactions between flexible and rigid segments result in loosely packed, non-crystalline hard domains with high chain mobility, leading to efficient energy dissipation during stretching and the oriented alignment of soft segments with reversible SIC behavior within the PU-AD_1Py_1 network. In addition to being simultaneously durable and robust, high-density H-bonding interactions in the hard domain with high chain mobility are expected to impart efficient self-healing capability for the PU-AD_1Py_1 network. For conventional thermoplastics, deformation or welding can occur due to the lower viscosity when they are exposed to temperatures significantly higher than T_g. However, the flow of thermoplastics should not normally be termed as self-healing because the networks are limited by molecular chain mobility and lack of reversible interactions at ambient temperature [42]. To validate the self-healing origin, in situ FT-IR spectroscopy was used to verify the dissociation and association ability of the reversible bonds under ambient conditions. As presented in Fig. 5 (b and c), as the temperature increases from 25 to 150 °C, the infrared absorption of free C=O groups in the PU-AD_1Py_1 networks gradually increases, whereas the peak height of C=O groups (ordered H-bonds, urethane, urea, and amide) clearly decreases. Moreover, the absorption intensities of the N–H stretching vibration and N–H bending decrease, illustrating the dissociation of H-bonds [18]. Furthermore, the intensity of each band is restored to its initial position, demonstrating that the H-bonds are reformed. Fig. S21 shows the corresponding peak intensities of free/ordered hydrogen-bonded carbonyl groups for PU-AD_1Py_1 as a function of temperature.

These results indicate the dissociation and associative ability of H-bonds upon changes in ambient conditions within hard domains of the dynamic PU-AD_1Py_1 network. As a block copolymer, the intrinsic self-healing of PU-AD_1Py_1 originates from its microphase-separated structure. Being above its glass transition temperature, the continuous soft phase enables polymer chain interdiffusion across the damaged interface, coupled with the simultaneous rearrangement of reversible hydrogen bonds with hard domains, ensuring efficient self-healing capability of the network. A self-healing mechanism is proposed in Fig. 5(a). Mechanical damage to PU-AD_1Py_1 networks causes chain cleavage and/or slippage, resulting in the subsequent formation of active groups exposed to the

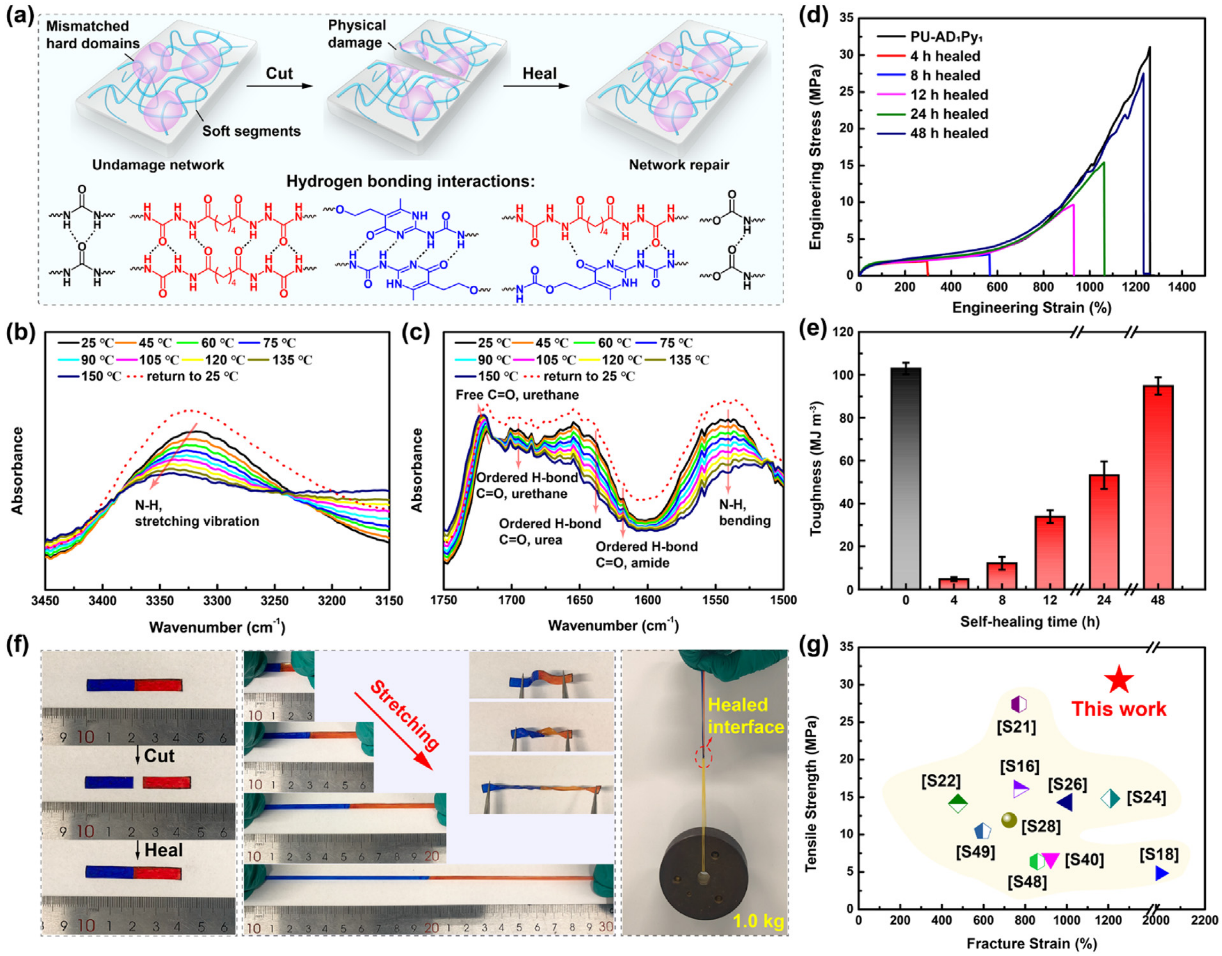

Fig. 5. Self-healing properties of PU-AD$_1$Py$_1$. (a) Schematic depiction of damage and reconstitution in dynamic PU-AD$_1$Py$_1$ networks. Temperature-dependent infrared spectra of PU-AD$_1$Py$_1$ samples collected from 25 to 150 °C: (b) 3150–3450 cm^{-1} and (c) 1500–1750 cm^{-1}. (d) Stress-strain curves and (e) corresponding toughness recoveries of PU-AD$_1$Py$_1$ films. (f) Photographs displaying the healing performance of the cutoff PU-AD$_1$Py$_1$ sample and the stretching, bending, twisting, and loading weight images of the repaired film. (g) Comparison of the mechanical performance of various room-temperature self-healing polymeric materials under similar healing conditions.

fracture interface. These active groups, including urea, urethane, and amine H-bonding interactions, are derived mainly from hard domains. Cleaved and/or displaced macromolecular segments undergo segmental arrangement, leading to conformational changes or diffusion at the fractured interface, and the damaged surface gradually heals with time [52]. Notably, SIC behavior does not directly drive the self-healing process. SIC behavior is mainly concentrated in soft segments and is responsible for toughening the PU-AD$_1$Py$_1$ network. The dynamic hydrogen bonding interactions between mismatched supramolecular segments in hard domains and chain mobility promote self-healing in the fracture interface. SIC behavior and self-healing properties of PU-AD$_1$Py$_1$ network are mutually compatible in the network structure design and jointly construct the high-toughness and room-temperature self-healing Supra-PU elastomers. To explore the healing ability, as demonstrated in Fig. 5(f), the cutoff PU-AD$_1$Py$_1$ samples are reconnected tightly to facilitate broken network reconstitution at room temperature. Upon contact of the fractured surfaces, hydrogen bonding reformation and chain diffusion promote interfacial re-bonding and restore mechanical integrity [12]. After being connected for 48 h, the healed film could withstand bending, twisting, stretching, and a 1.0 kg load without breaking (Fig. 5(f), Movie S1), demonstrating the excellent self-healing performance of the PU-AD$_1$Py$_1$ elastomers. To quantitatively evaluate the healing properties, tensile stress-strain tests were performed on the original and healed samples. As presented in Fig. 5(d), the broken network gradually recombines with healing for different time periods, and the stress-strain curve of the healed PU-AD$_1$Py$_1$ reaches a recovery tensile strength of 27.5 MPa and a fracture strain of 1233.4 % after healing for 48 h at room temperature. The mechanical healing efficiency (η) of PU-AD$_1$Py$_1$ can reach 92.1 %, with a restored toughness of 94.8 MJ m^{-3} (Fig. 5(e)), which is much greater than that of most reported room-temperature self-healing materials (Fig. 5(g), Table S4). As shown in Fig. S22, we further assessed the self-healing capacity through measurements of the repaired tensile strength and fracture strain, which reach restoration levels of 27.3 MPa and 1224.8 %, respectively. The corresponding healing efficiencies for tensile strength and fracture strain are as high as 89.2 % and 97.8 %, respectively.

2.6. Study of the luminescent fiber

Luminescent textiles woven from 1D luminescent fibers exhibit attractive advantages in terms of wearability and adaptability with various arbitrary shapes and curved surfaces, and are a new focus of attention as ideal wearable integration platforms [7]. Benefiting from its high mechanical performance and self-healing ability, PU-AD$_1$Py$_1$ has been employed to develop luminescent fibers. As displayed in Fig. 6(a), copper-doped zinc sulfide microparticles (Cu-ZnS) were dispersed homogenously into the PU-AD$_1$Py$_1$ resin mixture to fabricate the spinning solution. The intrinsic semiconductor Cu-ZnS phosphors exhibit an irregular oval shape with an average

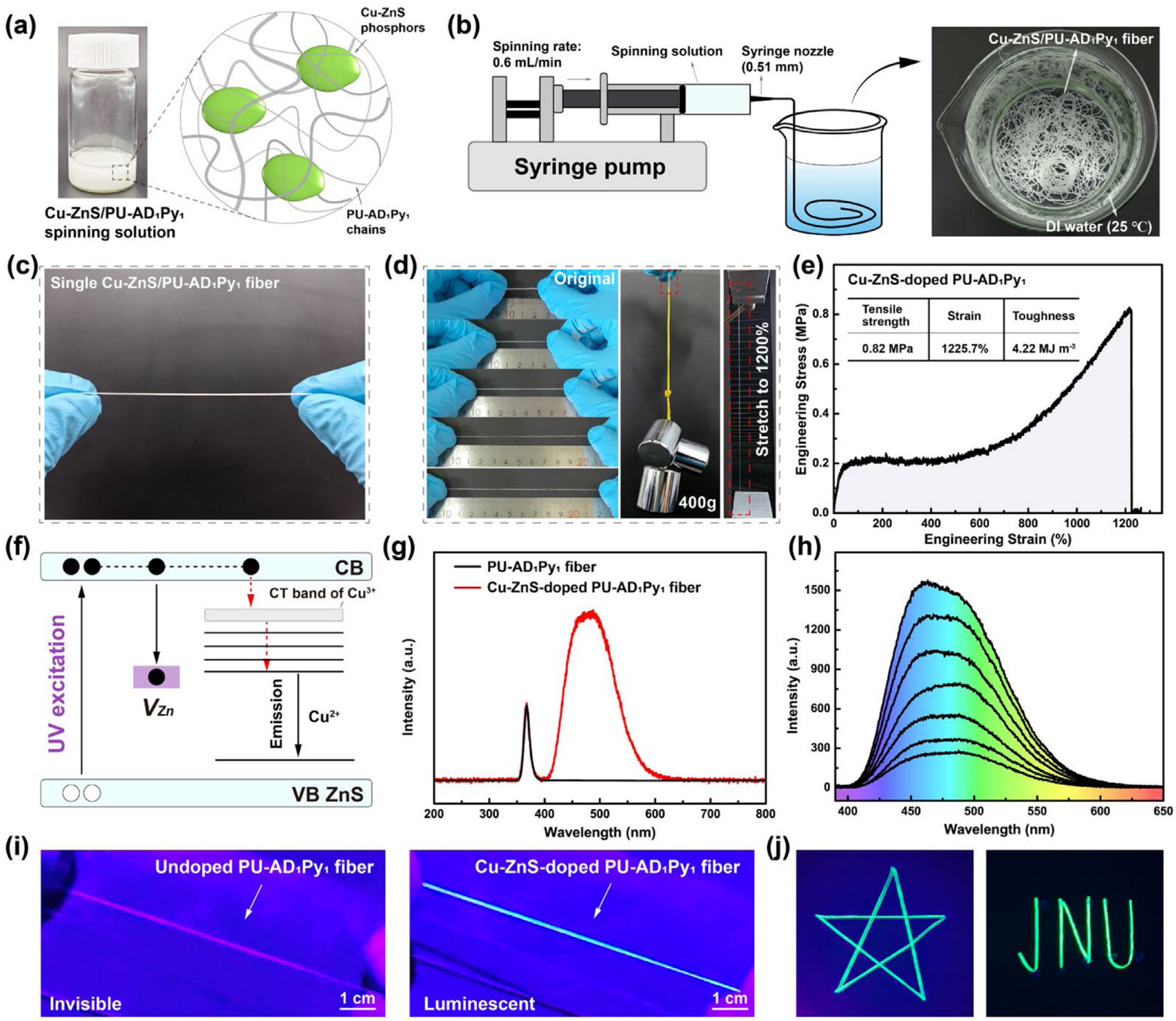

Fig. 6. (a) Schematic of the Cu-ZnS/PU-AD_1Py_1 network structure. (b) Fabrication procedures and photographs of the collected luminescent fibers. (c) Macroscopic view of a single luminescent fiber. (d) Stretching and loading weight (400 g) photos of a single luminescent fiber. (e) Tensile stress-strain curve of the luminescent fiber and the inset table showing the mechanical parameters. (f) Photoluminescence (PL) mechanism of copper-doped zinc sulfide phosphors. (g) PL spectra of PU-AD_1Py_1 and Cu-ZnS-doped PU-AD_1Py_1 fibers. (h) Afterglow spectra of luminescent fibers with different delay times. (i) Photos of original and luminescent fibers. (j) Fluorescence images with various patterns of luminescent fibers.

particle size of approximately 20 μm (Fig. S23(a)), and energy dispersive X-ray spectroscopy (EDX) spectral mapping reveals the elemental distribution of the Cu-ZnS particles, including the elements S, Cu, and Zn (Fig. S23(b–d)). A small Zn Kα peak (~8.04 keV) is observed in the EDX spectrum, which illustrates the existence of Cu in the phosphors (Fig. S24). XRD confirms the wurtzite structure (JCPDS no. 36–1450) of ZnS (Fig. S25) [6]. Then, the spinning solution was injected into the deionized water bath via a nozzle with an inner diameter of 0.65 mm, and continuous luminescent fibers were collected after solvent exchange (Fig. 6(b)).

The corresponding macroscopic view of a single luminescent fiber is displayed in Fig. 6(c). The mechanical performance of the Cu-ZnS-doped PU-AD_1Py_1 luminescent fiber was characterized via uniaxial tensile testing. A J-shaped stress-strain curve with typical parameters, including tensile strength, fracture strain, and toughness, was obtained at a stretching rate of 50 mm min^{-1} (Fig. 6(e)). As shown in Fig. 6(d), the Cu-ZnS-doped PU-AD_1Py_1 fiber can withstand varying degrees of tensile deformation, with a maximum elongation-at-break of ~1200 %. Such superior mechanical robustness is further displayed by the fact that the single Cu-ZnS-doped PU-AD_1Py_1 fiber (0.01 g) could fit ~40,000 times its own weight (400 g) after folding. Moreover, the Cu-ZnS nanoparticles luminesce when the PL material is subjected to ultraviolet excitation. An opinion about the PL mechanism is proposed in Fig. 6(f). Under the ultraviolet excitation, most of the excited electrons transfer to the excited level of Cu^{2+}, and the remaining electrons then transfer to the energy level of Zn, producing blue-green luminescence through the recombination of electrons and holes. Compared with the PU-AD_1Py_1 fiber, the Cu-ZnS-doped PU-AD_1Py_1 fiber exhibits bright blue-green emission when irradiated with ultraviolet light (365 nm). A broad band from 400 to 650 nm is found, which may correspond to zinc vacancy (V_{Zn}) emission and the transition from the defect level to the copper-induced t_2 level (Fig. 6(g and i)) [53]. To evaluate its self-healing properties, the cutoff Cu-ZnS-doped PU-AD_1Py_1 fiber was reconnected and healed for 48 h at ambient temperature. As shown in Fig. S26, the disrupted luminescent network can heal following the dynamic reconstruction of the self-healing elastomer. The Cu-ZnS-doped PU-AD_1Py_1 fiber recovers to its initial luminescent state with nearly unvaried emission luminescent intensity and distribution under the 365 nm UV irradiation. Meanwhile, the stress-strain curve of the healed Cu-ZnS-doped PU-AD_1Py_1 fiber reaches a recovery tensile strength of 0.75 MPa and a fracture strain of 1164.2 %. Intriguingly, the cyan afterglow at approximately 480 nm could last $\sim$ 0.8 s after ceasing the 365 nm UV irradiation (Figs. 6(h) and S27). This process involves the generation of electrons and holes upon photoexcitation, charge capture, release, and charge recombination. Owing to their superior mechanical and PL performance, various patterns assembled from the Cu-ZnS-doped PU-AD_1Py_1 fiber, such as the "pentagram" and "JUN" patterns, can be clearly observed in the dark (Fig. 6(j), Movies S2 and S3). Furthermore, we envision great potential for the luminescent fibers in wearable safety clothing for nighttime work. As a proof-of-concept, we demonstrate a T-shirt with an "SOS" pattern (Fig. S28), which could emit a macroscopic SOS signal when people are encountering danger at night with no available network signals.

3. Conclusions

This study demonstrates a spider-silk-inspired concept for achieving a supramolecular hydrogen-bonding crosslinked poly(urethane-urea) elastomer (Supra-PU) with extraordinary room-temperature self-healing ability and high mechanical performance, and further develops toughening luminescent fibers. Such H-bond interactions are beneficial for eliminating excessive supramolecular binding without sacrificing strength or resilience. Owing to the mismatched mechanical interplay between flexible/rigid supramolecular segments, PU-AD_1Py_1 elastomers present denser hard domains with effective energy dissipation and prominent SIC. As a result, the PU-AD_1Py_1 elastomers exhibit a high tensile strength of 30.6 MPa, ultrahigh fracture strain of 1251.8 %, superb toughness of 102.9 MJ m^{-3}, prominent notch insensitivity with high fracture energy of 4.7 kJ m^{-2}, and repeated deformation recovery. High-density H-bond interactions in hard domains render the desired network sacrificial and reversible, facilitating polymer chain interpenetration and fast reorganization. The broken PU-AD_1Py_1 film initiates the healing process at room temperature after being reconnected for 48 h, resulting in a high healing efficiency of 92.1 % with a recovered tensile strength of 27.5 MPa, a fracture strain of 1233.4 %, and a toughness of 94.8 MJ m^{-3}. Luminescent poly(urethane-urea) fibers were fabricated by incorporating Cu-ZnS phosphors into the poly(urethane-urea) resin via wet spinning, which exhibited satisfactory mechanical robustness and afterglow emission of $\sim$0.8 s. Furthermore, the scalable production of the poly(urethane-urea) luminescent fiber faces defined challenges but remains promising. The spinning demands precise control over coagulation bath parameters to achieve a uniform cross-section. And the polymer synthesis requires exact stoichiometry for reproducibility. Fortunately, the raw materials are commercially available and inexpensive, and the step-growth polymerization is a well-established method, making scaled production a realistic goal.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ting Ye: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Fang Zhang:** Validation, Software, Investigation. **Tao Zhang:** Visualization, Methodology, Investigation. **Liyang Song:** Investigation, Formal analysis. **Jinglei Tang:** Validation, Software. **Chen Zhang:** Visualization, Investigation. **Yunjie Yin:** Writing – review & editing, Resources, Methodology, Investigation. **Katja Loos:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization. **Chaoxia Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jmst.2025.10.075.

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